

Network Science — Chapter 2: Graph Theory

Formula Sheet — All Symbols Defined

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1 LINKS & DEGREES — UNDIRECTED

Total number of links:

$$L = \frac{1}{2} \sum_{i=1}^N k_i$$

L total number of links

k_i degree of node i (number of its links)

$\frac{1}{2}$ each link touches two nodes, so summing all k_i double-counts

Average degree:

$$\langle k \rangle = \frac{1}{N} \sum_{i=1}^N k_i = \frac{2L}{N}$$

$\langle k \rangle$ mean degree across all nodes

N total number of nodes

$2L$ factor of 2 because each link contributes to two degrees

Maximum possible links (complete graph):

$$L_{\max} = \binom{N}{2} = \frac{N(N-1)}{2}$$

$L_{\max}$ max links; achieved only in a complete graph

$\binom{N}{2}$ ways to choose 2 nodes from N

A network is **sparse** if $L \ll L_{\max}$.

2 LINKS & DEGREES — DIRECTED

Total degree of a node:

$$k_i = k_i^{\text{in}} + k_i^{\text{out}}$$

k_i^{in} in-degree: links *pointing to* node i

k_i^{out} out-degree: links *leaving* node i

Total directed links:

$$L = \sum_{i=1}^N k_i^{\text{in}} = \sum_{i=1}^N k_i^{\text{out}}$$

L each directed link contributes 1 to one in-degree and one out-degree (no double-counting, no $\frac{1}{2}$)

Average degree (directed):

$$\langle k^{\text{in}} \rangle = \langle k^{\text{out}} \rangle = \frac{L}{N}$$

$\langle k^{\text{in}} \rangle$ mean in-degree

$\langle k^{\text{out}} \rangle$ mean out-degree (always equal)

L/N note: L/N not $2L/N$ — no shared endpoints

3 DEGREE DISTRIBUTION

Degree distribution:

$$p_k = \frac{N_k}{N}$$

p_k prob. a randomly chosen node has degree k

N_k number of nodes with degree exactly k

N total nodes

Normalisation:

$$\sum_{k=0}^{\infty} p_k = 1$$

Every node must have *some* degree; probabilities sum to 1.

Average degree via distribution:

$$\langle k \rangle = \sum_{k=0}^{\infty} k p_k$$

k a specific degree value

p_k probability of that degree

Standard mean of a discrete distribution.

4 ADJACENCY MATRIX

Degree from matrix (undirected):

$$k_i = \sum_{j=1}^N A_{ij} = \sum_{j=1}^N A_{ji}$$

A_{ij} 1 if link between i and j , else 0

$A_{ij} = A_{ji}$ matrix is symmetric for undirected graphs

In/out-degree from matrix (directed):

$$k_i^{\text{in}} = \sum_{j=1}^N A_{ij}, \quad k_i^{\text{out}} = \sum_{j=1}^N A_{ji}$$

A_{ij} 1 if a link points *from* j *to* i , else 0

Link count from matrix:

$$2L = \sum_{i,j} A_{ij} \quad (\text{undirected})$$

$2L$ each undirected link appears as both A_{ij} and A_{ji}

Weighted network:

$$A_{ij} = w_{ij}$$

w_{ij} weight of link between i and j (any ≥ 0)

Node strength (weighted analogue of degree): $s_i = \sum_j w_{ij}$

5 PATHS & DISTANCES

Number of paths of length d :

$$N_{ij}^{(d)} = [A^d]_{ij}$$

$N_{ij}^{(d)}$ number of paths of exactly d steps from i to j

A^d adjacency matrix raised to the d -th power. Shortest path $d_{ij} =$

$[A^d]_{ij}$ entry of A^d

smallest d for which $[A^d]_{ij} > 0$.

Average path length (directed):

$$\langle d \rangle = \frac{1}{N(N-1)} \sum_{\substack{i,j=1 \\ i \neq j}}^N d_{ij}$$

d_{ij} shortest path distance between i and j

$N(N-1)$ total ordered pairs (i, j) with $i \neq j$

For undirected: $\langle d \rangle = \frac{1}{\frac{N(N-1)}{2}} \sum_{i < j} d_{ij}$

Computed only within connected components.

6 CLUSTERING COEFFICIENT

Local clustering coefficient:

$$C_i = \frac{2L_i}{k_i(k_i - 1)}$$

C_i clustering coeff. of node i ($0 \leq C_i \leq 1$)

L_i actual links among i 's k_i neighbours

$k_i(k_i - 1)/2$ max possible links among k_i neighbours

$C_i = 0$ no neighbours connected to each other

$C_i = 1$ all neighbours form a clique

0) when $k_i \leq 1$.

Average clustering coefficient:

$$\langle C \rangle = \frac{1}{N} \sum_{i=1}^N C_i$$

$\langle C \rangle$ prob. that two neighbours of a random node are also connected

Global clustering coefficient:

$$C_{\Delta} = \frac{3 \times (\text{number of triangles})}{(\text{number of connected triplets})}$$

C_{Δ} weights each triplet equally (not each node)

triangle 3 nodes all mutually connected

triplet 3 nodes A-B-C where A-B and B-C exist $\langle C \rangle$ weights nodes

factor 3 each triangle contains 3 triplets

equally; C_{Δ} weights triplets equally — they differ.

7 STATISTICAL QUANTITIES

Mean:

$$\langle x \rangle = \frac{1}{N} \sum_{i=1}^N x_i$$

n -th moment:

$$\langle x^n \rangle = \frac{1}{N} \sum_{i=1}^N x_i^n$$

Standard deviation:

$$\sigma_x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2}$$

σ_x spread of x values around the mean

$\langle x^2 \rangle - \langle x \rangle^2$ variance

8 METCALFE'S LAW

$$\text{Value} \sim N^2, \quad \text{Cost} \sim N$$

N^2 approximate number of pairwise connections

Network becomes prof-

N cost grows linearly with users

itable when $N^2 \gg N$, i.e. for large N . **Caveat:** real networks are sparse ($L \ll N^2/2$), so actual value $\sim N$, not N^2 .

GRAPH TYPE COMPARISON

Graph type	Links L	Avg degree $\langle k \rangle$	A_{ii}	$A_{ij} = A_{ji}?$
Undirected	$\frac{1}{2} \sum_{ij} A_{ij}$	$2L/N$	$= 0$	Yes
Undirected + self-loops	$\frac{1}{2} \sum_{i \neq j} A_{ij} + \sum_i A_{ii}$	*	$\neq 0$	Yes
Multigraph	$\frac{1}{2} \sum_{ij} A_{ij}$	$2L/N$	$= 0$	Yes
Directed	$\sum_{ij} A_{ij}$	L/N	$= 0$	No
Weighted undirected	$\frac{1}{2} \sum_{ij} \mathbf{1}[A_{ij} > 0]$	$2L/N$	$= 0$	Yes
Complete graph	$N(N-1)/2$	$N-1$	$= 0$	Yes

* Depends on whether a self-loop counts as 1 or 2 towards degree (ambiguous by convention).

COMPLETE FORMULA REFERENCE TABLE

#	Quantity	Formula	Symbol meanings
1	Links (undirected)	$L = \frac{1}{2} \sum_i k_i$	k_i : degree of node i ; factor $\frac{1}{2}$ removes double-count
2	Avg degree (undirected)	$\langle k \rangle = 2L/N$	N : total nodes
3	Total degree (directed)	$k_i = k_i^{\text{in}} + k_i^{\text{out}}$	in-degree + out-degree
4	Links (directed)	$L = \sum_i k_i^{\text{in}} = \sum_i k_i^{\text{out}}$	no $\frac{1}{2}$ factor (no double-count)
5	Avg degree (directed)	$\langle k^{\text{in}} \rangle = \langle k^{\text{out}} \rangle = L/N$	always equal to each other
6	Degree distribution	$p_k = N_k/N$	N_k : nodes with degree k
7	Normalisation	$\sum_k p_k = 1$	probabilities sum to 1
8	Avg degree via p_k	$\langle k \rangle = \sum_k k p_k$	mean of discrete distribution
9	Degree from A (undirected)	$k_i = \sum_j A_{ij}$	row/column sum of symmetric A
10	In/out-degree from A	$k_i^{\text{in}} = \sum_j A_{ij}; k_i^{\text{out}} = \sum_j A_{ji}$	row sum = in; col sum = out
11	Non-zero elements of A	$\sum_{ij} A_{ij} = 2L$	each undirected link appears twice in A
12	Max links	$L_{\text{max}} = N(N-1)/2$	complete graph; sparsity if $L \ll L_{\text{max}}$
13	Weighted adjacency	$A_{ij} = w_{ij}$	$w_{ij} \geq 0$; node strength $s_i = \sum_j w_{ij}$
14	Paths of length d	$N_{ij}^{(d)} = [A^d]_{ij}$	(i, j) entry of A raised to power d
15	Avg path length	$\langle d \rangle = \frac{1}{N(N-1)} \sum_{i \neq j} d_{ij}$	d_{ij} : shortest path between i and j
16	Local clustering coeff.	$C_i = \frac{2L_i}{k_i(k_i-1)}$	L_i : links among i 's neighbours
17	Avg clustering coeff.	$\langle C \rangle = \frac{1}{N} \sum_i C_i$	node-weighted average
18	Global clustering coeff.	$C_{\Delta} = \frac{3 \times \text{triangles}}{\text{triplets}}$	triplet-weighted; differs from $\langle C \rangle$